EDDQC: Enhanced Dynamical Distributing Quantum Compilation

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***Abstract*—This article presents enhanced dynamical distributing quantum compilation (EDDQC), an optimized method for distributed quantum computing (DQC) using linear nearest neighbor (LNN) architecture integrated with quantum switches. By leveraging the symmetry of LNN topology and designating dangling qubits as communication links, our approach optimizes compilation for high local connectivity, sparse full connectivity algorithms (HLC-SFC) like quantum approximate optimization algorithm (QAOA) and quantum Fourier transform (QFT). Experimental results demonstrate significant performance improvements over traditional methods, including reductions in cross-group swaps by up to 67.2%, gate count by 43.8%, and total execution cycles by up to 40%. We also utilize the area law of entanglement entropy to limit entanglement growth in our 1-D system. Our comprehensive approach combines LNN chains' efficiency with reconfigurable network flexibility, enhancing scalability and robustness for large-scale quantum computations.**

***Index Terms*—Dangling qubit, distributed quantum computing (DQC), linear compiler, linear nearest neighbor (LNN), quantum switch.**

I. Introduction

QUANTUM computing promises faster solutions in fields, such as factorization [1], finance [2], and chemistry [3]. As quantum computing progresses from the noisy intermediate-scale quantum (NISQ) era [4], [5] toward fault-tolerant quantum computing (FTQC) [6], [7], many applications demand more resources than a single device can provide. Distributed quantum computing (DQC) offers a way to scale quantum systems by utilizing multiple processors [8], [9].

In the DQC framework, each quantum computer functions as an independent quantum processing unit (QPU), with quantum teleportation facilitating communication between different QPUs [8]. Quantum teleportation, implemented using nonlocal two-qubit gates, is highly susceptible to system coherence times and noise, which represent two of the primary challenges in DQC systems. In addition, conventional compilation in heavy-hex topology introduces an excess number of swap gates, leading to increased compilation overhead, deeper circuits, and higher errors [10].

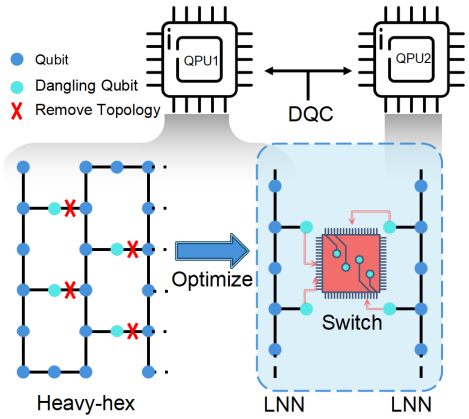


Fig. 1. Proposed chip-to-chip distributed quantum system by modifying the heavy-hex topology to LNN topology and taking the dangling qubits as a teleportation pair.

To address these challenges, as illustrated in Fig. 1, we restructure the heavy-hex topology into a linear nearest neighbor (LNN) topology, removing specific qubit connections to reduce compilation overhead [11]. This transformation results in dangling qubits within the LNN structure, which can serve as teleportation qubits, allowing for the application of nonlocal gates to facilitate communication between QPUs.

Building upon this foundation, we further enhance our approach by integrating quantum switches to collaborate with our dangling-qubits approach to form an optimized interconnect topology for DQC. These quantum switches function as entanglement resource managers, generating and distributing Bell pairs between QPUs using photonic channels. Each switch can dynamically reconfigure the pairs of qubits that interact by establishing quantum connections between multiple communication qubits, allowing the network to adapt to the computational requirements of specific algorithms.

For quantum chips with the same heavy-hex topology, the topological transformation will result in the same LNN structure with dangling qubits. This symmetry structure allows dangling qubits to communicate with each other one-to-one, thus deciding the structure of the DQC system. This structure also allows DQC to use LNN compilation directly and reduces noise and circuit depth, which makes the DQC system more

Received 30 April 2025; revised 11 July 2025 and 7 August 2025; accepted 14 August 2025. Date of publication 29 August 2025; date of current version 30 December 2025. This work was supported in part by the IBM-RPI Future of Computing Research Collaboration and in part by the U.S. National Science Foundation under Award 2519029. *(Corresponding author: Zhiding Liang.)*

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Digital Object Identifier 10.1109/TVLSI.2025.3600096

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reliable and practical. Our approach optimizes quantum circuit compilation in several key ways.

1) *Efficient LNN Transformation on DQC*: We restructure the heavy-hex topology into an LNN architecture for DQC structure and minimize nonlocal interactions between chips and reduce SWAP gates on each chip, particularly benefiting high local connectivity, sparse full connectivity (HLC-SFC) algorithms like quantum approximate optimization algorithm (QAOA) and quantum Fourier transform (QFT).
2) *Strategic Dangling-Qubit Utilization*: We designate dangling qubits at the ends of LNN chains as communication links, leveraging their natural position to minimize disruption to computational qubits.
3) *Quantum-Switch Integration*: By connecting dangling qubits across QPUs through quantum switches, we enable dynamic reconfiguration of communication paths, optimizing entanglement distribution and reducing cross-group operations.
4) *Area Law Exploitation*: We leverage the favorable entanglement scaling properties of 1-D systems to maintain manageable entanglement growth as the system size increases.
5) *Performance-Driven Resource Allocation*: Our framework dynamically evaluates the cost of qubit movement versus quantum-switch utilization, making optimal decisions that minimize both time and error accumulation.

Through comprehensive experimentation, we demonstrate that our enhanced dynamical distributing quantum compilation (EDDQC) approach significantly outperforms both traditional compilation methods and standard LNN-based DQC [12]. For QAOA circuits, we achieve up to a 35% reduction in cross-group swaps, 20% lower gate count, and 40% fewer total execution cycles compared with DQC with single teleportation. QFT circuits show similar improvements, with up to a 45% reduction in cross-group swaps and a 25% decrease in gate count. The remainder of this article is organized as follows. Section II provides background information on quantum teleportation, DQC, linear nearest-neighbor architecture, and quantum switches. Section III discusses our motivation for including quantum switches consideration further than LNN topology. Section IV presents our enhanced methodology, including a theoretical approach to proving the entanglement entropy of the LNN architecture, the physical integration of dangling qubits between QPUs, and our performance-driven search algorithm with its corresponding metrics and cost function. Section V details our experimental evaluation across multiple dimensions. Finally, Section VI concludes this article and discusses future directions.

II. Background

A. Linear Nearest-Neighbor Architecture in DQC

LNN architecture addresses scalability and compilation complexity challenges in DQC [13]. The symmetry of LNN enables a balanced qubit distribution, minimizing overhead during gate execution, and results in a reduced qubit movement complexity [14]. By transforming conventional heavy-hex topology into LNN, we leverage sparse connectivity and symmetry, significantly reducing nonlocal interactions and costly SWAP gates.

This architecture proves especially advantageous for HLC-SFC algorithms, including QAOA [15], [16], [17], and QFT [18], [19], [20], which benefit from minimal long-range connectivity. HLC-SFC algorithms represent a class of quantum algorithms that exhibit predominantly local qubit interactions with occasional long-range connectivity requirements. These algorithms are characterized by: 1) dense local gate operations within small neighborhoods of qubits; 2) sparse global connectivity patterns that can be efficiently decomposed into local operations and limited nonlocal interactions; and 3) natural compatibility with linear and near-linear topologies. QAOA exemplifies this pattern through its alternating local problem Hamiltonians and sparse mixing operations, while QFT demonstrates sparse full connectivity through its structured gate pattern that requires fewer long-range interactions when compiled to LNN architectures. The LNN chain's natural compatibility with existing heavy-hex lattice designs [21] allows qubits to interact directly with immediate neighbors, simplifying quantum gate implementation and reducing time costs for complex routing operations. This simplification leads to lower compilation overhead, accelerating the compilation process and enhancing computation reliability by minimizing potential error sources such as crosstalk.

B. Quantum Teleportation

Quantum teleportation enables quantum state transfer between locations without physical qubit transmission, utilizing quantum entanglement and classical communication [22]. In the standard protocol, a qubit state $|\phi\rangle = \alpha|0\rangle + \beta|1\rangle$ is teleported using a shared Bell state $\left|\Phi^+\right\rangle = (1/(2)^{1/2})(|00\rangle + |11\rangle)$, Bell measurement, and classical communication [23], [24]. In DQC, teleportation divides into two categories [8].

1) State Teleportation: State teleportation transmits quantum states at the cost of one entangled bit [25], enabling nonlocal operations by transferring qubit states between QPUs and executing operations locally.

2) Gate Teleportation: Gate teleportation uses one e-bit to execute two-qubit controlled operations without physical qubit transfer [26]. It entangles a control qubit with a communication qubit in a separate QPU, with both qubits remaining in their original locations [27]. Recent research shows multiple two-qubit operations can be teleported without additional cost when using shared control qubits [28], [29], [30].

C. Chip-to-Chip Distribution and Quantum-Switch Integration

DQC chip-to-chip distribution enables computation across interconnected quantum processors, ideal for applications requiring large qubit counts, such as quantum simulation, optimization, and quantum machine learning [31]. Each quantum chip maintains independent qubits and gates, collaborating through teleportation and state manipulation. Dangling qubits in the linear compiler facilitate state transmission between

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chips, requiring efficient quantum error correction to maintain information integrity [32].

The quantum switches [33], [34], [35], [36] bridge entanglement between bipartite qubit sets. Similar to classical network switches determining input–output interconnections, quantum switches manage entanglement distribution between QPUs. The model distinguishes between data qubits and communication qubits. This configuration connects through quantum switches functioning as entanglement resource managers, generating and distributing Bell pairs between QPUs via photonic channels. Each switch can establish connections between multiple communication qubits sequentially through Bell pair generation and quantum teleportation.

III. Motivations

Our research is motivated by the scalability challenges in quantum computing that become increasingly pronounced as qubit counts grow. Examining IBM's heavy-hex architecture revealed an opportunity in its structural properties that we leverage for DQC. The heavy-hex topology, when transformed into an LNN arrangement, creates distinct "dangling qubits" at chain endpoints that typically maintain minimal connections within their processor [37], [38], [39]. These dangling qubits represent optimal candidates for interprocessor communication without disrupting internal computational structures.

This observation formed the foundation for our chip-to-chip distributed compilation approach. The LNN architecture's symmetry inherently provides balanced qubit distribution, making it naturally suited for distribution across multiple processors while maintaining computational efficiency [13], [14]. By designating dangling qubits as communication interfaces, we minimize the overhead typically associated with distributed quantum computation, particularly for algorithms with predominantly local connectivity requirements such as QAOA and QFT [16], [19].

While pure LNN architectures would be theoretically optimal, no commercial quantum hardware implements such designs today. Our heavy-hex transformation strategy enables immediate deployment on existing IBM infrastructure without requiring new chip fabrication. This approach bridges the critical gap between theoretical optimization and practical implementation, providing quantum computing centers with a clear migration path to distributed computing using current technology. Rather than waiting for future hardware generations, our method delivers near-optimal DQC performance on today's systems.

Building on this chip-to-chip framework, we incorporated insights from quantum-switch technology to enhance connectivity flexibility. Recent experimental demonstrations of quantum switches [33], [36] have shown their capacity to establish high-fidelity entanglement between remote qubits dynamically. Unlike static interconnect topologies that impose fixed communication patterns, quantum switches create on-demand entanglement connections based on computational requirements, enabling adaptive resource allocation [35]. This dynamic reconfigurability represents a significant advancement over traditional DQC architectures. By integrating quantum switches with our dangling-qubit communication model, we create a system that maintains the computational benefits of the LNN topology within each processor while supporting efficient interprocessor communication through optimized entanglement distribution. The resulting architecture adapts to varying algorithm requirements without physical reconfiguration, reducing communication overhead and improving operational fidelity compared with conventional approaches [34].

Our approach, thus, combines the structural advantages of heavy-hex derived LNN compilation with the flexibility of quantum-switch technology, creating a scalable foundation for DQC that addresses both current hardware limitations and future computational demands.

IV. Methodology

In this section, we present our enhanced methodology that combines the benefits of LNN architecture with quantum-switch integration. We begin by examining the entanglement entropy considerations in LNN topology that provide theoretical justification for our approach. Next, we discuss our performance metrics and cost function optimization that account for both physical movement and quantum teleportation operations. Finally, we introduce our performance-driven searching algorithm that enables efficient orchestration of qubit operations in a distributed environment.

A. Entanglement Entropy in LNN Topology

In the LNN topology, qubits are arranged in a 1-D chain where each qubit interacts only with its immediate neighbors. This arrangement has significant implications for the entanglement properties of the system.

The area law states that for quantum systems with local interactions, the entanglement entropy of a subsystem scales with the size of its boundary rather than its volume [40]. For a d-dimensional system, the entanglement entropy $S(\rho_A)$ of a subsystem A is proportional to the boundary area $|\partial A|$

$$S(\rho_A) \propto |\partial A| \sim O\left(N^{1-1/d}\right)$$

where N is the total number of qubits and d is the dimensionality of the system.

In a 1-D system ($d = 1$), the boundary area is constant ($|\partial A| = 2$). This means that the entanglement entropy does not scale with the size of the subsystem A.

1) Mathematical Derivation in LNN Systems: Consider a 1-D LNN chain of N qubits. Let A be a contiguous block of N_A qubits. The boundary $|\partial A|$ of A remains constant: $|\partial A| = 2$. The entanglement entropy $S(\rho_A)$ is bounded by the number of interactions crossing the boundary

$$S(\rho_A) \leq c \cdot |\partial A|$$

where c is a constant depending on the specifics of the system. Substituting $|\partial A| = 2$, we obtain: $(\rho_A) \leq 2c$. This result implies that the entanglement entropy is independent of the size N_A of the subsystem A. Consequently, the system obeys the Area Law, making it suitable for efficient simulation since the amount of entanglement remains manageable. In terms of circuit depth and entanglement, in a quantum circuit of depth

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p, the entanglement can grow with each layer. However, in LNN, this growth is constrained by local interactions. The maximum entanglement entropy after p layers is $(\rho_A) \leq 2c \cdot p$.

This indicates that the entanglement entropy scales linearly with the circuit depth p, not with the system size N. Such scaling is advantageous for practical implementations, as it prevents exponential growth of entanglement, which would be detrimental to scalability. For example, in conformal field theory, the entanglement entropy scales logarithmically with the subsystem size

$$S(\rho_A) = \frac{c}{3} \log N_A + S_0$$

where c is the central charge of the theory.

2) Higher Dimensional Systems: In higher dimensional systems ($d \geq 2$), the boundary area increases with the subsystem size, leading to a different scaling behavior for the entanglement entropy

$$S(\rho_A) \propto N_A^{(d-1)/d}.$$

The entanglement entropy grows with the subsystem size N_A, and the specific scaling depends on the interaction properties and the topology of the system. This growth can pose challenges for simulation and implementation, as it may lead to significant entanglement across the system.

By leveraging the favorable entanglement scaling properties of LNN chains in our DQC approach, we ensure that quantum resources scale efficiently as system size increases, providing a solid theoretical foundation for our architecture.

B. Physical Integration of Dangling Qubits With Quantum Switches

The physical implementation of our EDDQC approach requires a careful integration of dangling qubits with quantum switches to enable efficient inter-QPU communication. This integration forms the foundation of our DQC architecture and determines its operational capabilities. Dangling qubits in our LNN topology are physically located at the endpoints of qubit chains, positioned at the periphery of each QPU. In the original heavy-hex architecture, these qubits maintain connections to only one or two other qubits within the same QPU, making them ideal candidates for external communication without disrupting the internal computational structure.

The physical connection between dangling qubits and quantum switches occurs through microwave resonators or optical interfaces, depending on the specific implementation technology. For superconducting qubit systems, such as those based on IBM's architecture, this connection typically involves coupling resonators that connect the dangling qubit to an interface component capable of generating and detecting photons. These interface components serve as transducers that convert between the microwave domain and the optical domain. Optical channels, typically fiber optic connections, then link the interface components to the quantum switch, enabling photonic communication between distant QPUs.

The quantum switch itself is physically implemented as a photonic device that can process quantum information encoded in photon states. It consists of photon sources and detectors for generating and detecting entangled photon pairs that mediate the interactions between dangling qubits. The switch also includes optical circuitry such as waveguides, beam splitters, and phase shifters that route photons and enable Bell-state measurements. In addition, a classical control system with electronics processes measurement results and coordinates the timing of operations across multiple QPUs. This approach resembles the shared quantum gate processing unit (S-QGPU) architecture [41], where centralized quantum resources facilitate remote gate operations between distributed nodes.

Recent experimental implementations of quantum switches have demonstrated the ability to establish entanglement between remote qubits with high fidelity. Research published in [36] has achieved entanglement fidelities exceeding 0.95 for superconducting qubits separated by distances of up to several meters, while work from [33] has shown successful photonic Bell-state measurements with success probabilities above 0.9.

The operational flow of quantum information through this integrated system proceeds as follows. The quantum switch generates entangled photon pairs and sends one photon to each of two target dangling qubits located on different QPUs. Each dangling qubit interacts with its received photon, transferring the entanglement from the photon pair to the qubit pair. A series of measurements confirms the successful establishment of entanglement between the dangling qubits. The entangled dangling qubits are now available for teleportation operations or as resources for nonlocal gates between the QPUs. Finally, the quantum switch can dynamically establish new entanglement connections between different pairs of dangling qubits based on the computational requirements.

This physical implementation enables our quantum switch-based DQC system to achieve the theoretical performance improvements demonstrated in our evaluation. By establishing direct entanglement connections between dangling qubits, we avoid the need for extensive physical qubit movements, reducing both the time and error costs associated with distributed quantum computations. The specific technical requirements for this implementation align with current research directions in quantum networking technologies, making our approach both theoretically sound and experimentally feasible in the near term. The modular nature of this architecture also allows for incremental scaling, where additional QPUs can be integrated into the network by connecting their dangling qubits to existing or new quantum switches.

C. Performance Metrics and Cost Function Optimization

To evaluate qubit movement and the utilization of quantum switches, we derived performance metrics based on IBM's latest quantum processor along with theoretical insights from relevant quantum switch and teleportation studies [35], [36]. Table I provides a summary of these operational metrics, encompassing time and error rates for various quantum operations, including teleportation-based operations.

Our metrics account for both traditional swap operations and quantum teleportation performance. State teleportation incurs slightly higher time costs and error rates compared with gate teleportation, reflecting the additional operations required to transfer a quantum state completely. The quantum-switch

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TABLE I

PERFORMANCE METRICS FOR QUANTUM OPERATIONS. VALUES MARKED WITH ∗ ARE DIRECTLY FROM IBM QUANTUM DOCUMENTATION. VALUES MARKED WITH † ARE DERIVED FROM THEORETICAL QUANTUM-SWITCH FRAMEWORKS [36]. VALUES MARKED WITH ‡ ARE CALCULATED BASED ON TELEPORTATION PROTOCOL ANALYSIS [25], [26], [27]

Operation	Time (μs)	Error Rate
Swap∗	0.235	0.007
Cross-Group Swap†	0.5	0.035
Move∗	0.1	0.003
Teleportation State‡	0.6	0.020
Teleportation Gate‡	0.5	0.018
Switch Interaction†	0.4	0.015
CX∗	0.235	0.007
RZ∗	0.02	0.0001

interactions show improved performance compared with cross-group swaps, as they leverage entanglement resources more efficiently without physically moving qubits.

Our cost function integrates these parameters, assigning an error weight to balance time and error rate contributions. This function assists the algorithm in efficiently orchestrating qubit operations, enhancing both performance and reliability [42].

The combined cost C for an operation is calculated as

$$C = T_{\text{adj}} + W_{\text{error}} \times E$$

where T_{adj} is the adjusted time cost accounting for the expected number of repeats due to errors, W_{error} is the error weight representing the importance of error rate relative to time, and E is the error rate of the operation. The adjusted time cost T_{adj} is calculated as

$$\begin{aligned} P(k) &= (1-E) \times E^{k-1} \\ E[X] &= \sum_{k=1}^{\infty} \left[k \times (1-E) \times E^{k-1}\right] \\ &= (1-E) \times \left(1/(1-E)^2\right) = 1/(1-E) \\ T_{\text{adj}} &= T \times (1/(1-E)) \end{aligned}$$

where X is the random variable representing the number of attempts until success, $P(k)$ represents the probability of success on attempt k, $E[X]$ is the expected number of repeats, T is the nominal time cost of the operation, and $1/(1-E)$ represents the expected number of repeats.

For quantum teleportation operations, we must account for both the entanglement and the classical communication. When evaluating teleportation costs, we consider

$$\begin{aligned} C_{\text{teleport}} = T_{\text{entangle}} + T_{\text{measure}} + T_{\text{classical}} \\ + W_{\text{error}} \times \left(E_{\text{entangle}} + E_{\text{measure}}\right) \end{aligned}$$

where T_{entangle} is the time to generate the entangled pair, T_{measure} is the time for Bell-state measurement, $T_{\text{classical}}$ is the classical communication time, and E_{entangle} and E_{measure} are the respective error rates.

The cost function is used for operation selection and path-finding. The algorithm calculates the combined cost for possible strategies like swapping versus using a quantum switch and selects the one with the lowest cost. When finding paths for qubit movement, the cost function helps in choosing the least expensive route, considering both time and error implications.

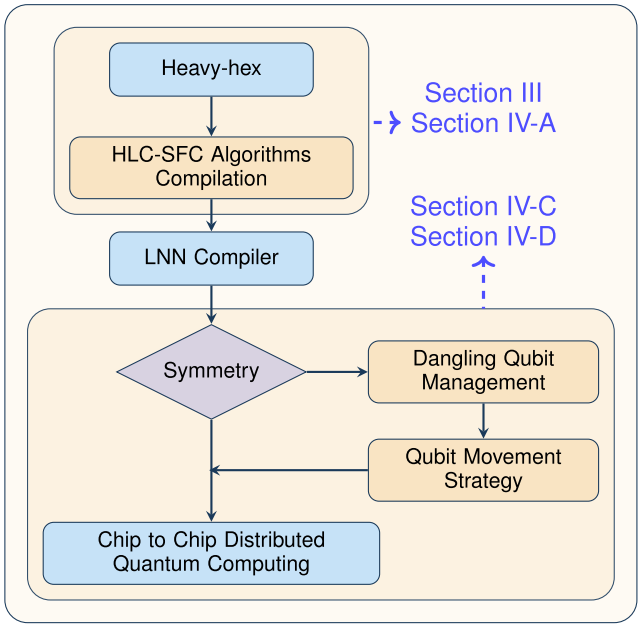


Fig. 2. Overview of the chip-to-chip DQC compilation process. The process begins with a heavy-hex architecture and compiles HLC-SFC quantum algorithms, such as QAOA and QFT, using an LNN compiler. The compiler leverages symmetry to optimize the structure, managing dangling qubits and employing a qubit movement strategy. This leads to an efficient chip-to-chip DQC system with minimized nonlocal interactions and enhanced scalability.

D. Performance-Driven Searching Algorithm

Our performance-driven searching algorithm is designed to optimize qubit interactions in a DQC environment using quantum switches. The core idea is to balance the tradeoff between physical qubit movement and direct interactions via quantum switches, ultimately reducing total execution costs and minimizing qubit decoherence due to movement (see Fig 3). This optimization is particularly crucial for large-scale quantum circuits where communication latency and error accumulation can degrade performance.

The algorithm is structured around key components.

1) Initialization and System Setup: The algorithm begins by initializing the qubit positions using the default IBM quantum processor layout without special ordering optimization, following the standard heavy-hex topology arrangement before transformation to LNN structure. The initial qubit mapping preserves the physical connectivity constraints of the target hardware while preparing for subsequent LNN transformation. We then construct an adjacency list that represents all potential movement and interaction paths. Each edge in the adjacency list corresponds to either a swap operation or a quantum-switch interaction, with associated costs derived from empirical performance metrics. These metrics account for the time and error rate of each operation, reflecting real-world performance constraints and hardware characteristics, such as those of IBM's latest superconducting qubit architectures.

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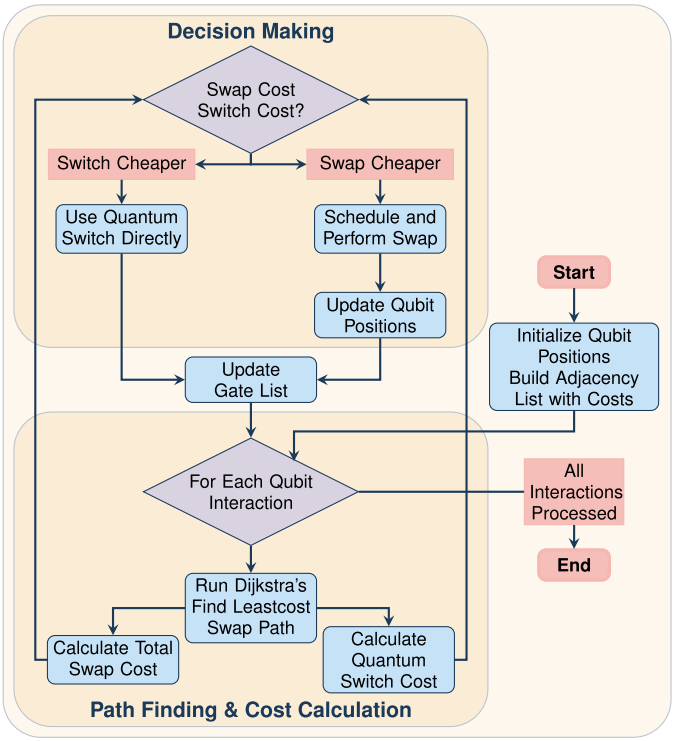


Fig. 3. Overview of the performance-driven searching algorithm steps.

2) Adjacency List Construction: Constructing the adjacency list is a fundamental step that defines the network of possible interactions. The list includes the following.

1) *Intragroup Swaps:* Movements within the same QPU, typically incurring lower costs.
2) *Cross-Group Swaps:* Movements between different groups of qubits are often more costly due to increased communication overhead.
3) *Quantum-Switch Interactions:* Direct, nonmovement-based interactions facilitated by quantum switches can provide substantial efficiency gains under the right conditions.
4) *Teleportation Operations:* State and gate teleportation operations that utilize entanglement resources to communicate between QPUs.

This adjacency list serves as the foundation for the algorithm's path optimization and cost analysis.

3) Path-Finding With Dijkstra's Algorithm: To determine the most cost-effective movement paths, the algorithm employs Dijkstra's algorithm, a classical approach to finding the shortest path in a weighted graph. In our context, "shortest" translates to "least cost," where cost is a composite measure of time and error likelihood. Unlike heuristic-based methods, such as A*, Dijkstra's algorithm is deterministic and guarantees an optimal path without assumptions, which is particularly valuable when dealing with critical quantum operations where precision is paramount.

4) Dynamic Cost Evaluation and Strategy Selection: The heart of the algorithm lies in its dynamic evaluation and decision-making process. For each qubit pair that needs to interact, the algorithm holds the following.

1) *Computes the Cost of Swapping:* Using the least-cost path from Dijkstra's algorithm, the total cost of performing the required swaps is calculated, accounting for any cross-group movements.
2) *Computes the Cost of Using a Quantum Switch:* The cost of performing a direct quantum-switch interaction is also computed, using the performance metrics.
3) *Computes the Cost of Teleportation:* The cost of performing either state or gate teleportation is calculated based on the specific operation needed.
4) *Compares All Strategies:* The algorithm then compares the costs and selects the most efficient strategy. This decision-making process ensures that qubit interactions are executed in a way that minimizes overall time and error accumulation.

5) Execution and System Updates: Once the strategy is chosen, the algorithm executes the operations accordingly. If swaps are selected, they are scheduled and performed in sequence, with the qubit positions updated to reflect their new locations. If a quantum-switch interaction or teleportation is chosen, it is executed directly, bypassing the need for physical qubit movement. The algorithm then updates the gate list to document all operations performed, ensuring an accurate representation of the quantum circuit's current state.

6) Adaptability and Scalability: One of the algorithm's strengths is its adaptability. By continuously evaluating the current system state and recalculating costs for each interaction, it remains responsive to the changing demands of the circuit. This adaptability is crucial for scaling the algorithm to handle complex circuits with hundreds of qubits, where efficient resource management can make the difference between a viable and impractical quantum computation.

In addition, the algorithm's design considers real-world constraints, such as varying qubit coherence times and the limitations of physical hardware. This practical awareness allows it to make informed decisions that align with the operational characteristics of current quantum processors. Algorithm 1 efficiently balances cost and operational overhead by dynamically selecting between qubit movement, quantum-switch interactions, and teleportation operations. Leveraging Dijkstra's algorithm for optimal path-finding and incorporating cost-effective decision-making ensures scalability for handling extensive qubit networks and complex interaction graphs.

V. Evaluation

A. Experiment Setting

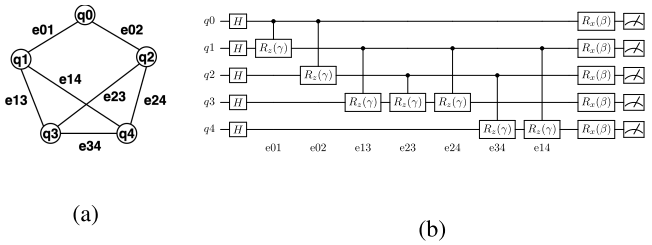
We evaluate our enhanced EDDQC method through a series of experiments designed to assess both the theoretical advantages of LNN-based DQC and the practical performance improvements gained through quantum-switch integration. Our evaluation framework is structured to progressively demonstrate the benefits of our approach, beginning with the baseline performance of LNN compilation and then examining the enhancements provided by quantum-switch integration.

The experiments target IBM's quantum architecture, utilizing both simulated environments and performance models based on real-hardware specifications. For the LNN compiler

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Algorithm 1 Performance-Driven Searching Algorithm

procedure SEARCHOPTIMALSTRATEGY(Q, P, M, S, Ω)
  Initialize qubit positions Q and build adjacency list P with costs based on metrics M
  for each qubit interaction $s \in S$ **do**
    $p \leftarrow$ DIJKSTRA(Q, P, s) ▷ Find least-cost swap path using Dijkstra's algorithm
    $c_{swap} \leftarrow$ Compute total cost of swaps along p
    $c_{switch} \leftarrow$ Compute cost of using quantum switch
    $c_{teleport} \leftarrow$ Compute cost of teleportation
    $c_{min} \leftarrow \min(c_{swap}, c_{switch}, c_{teleport})$
    if $c_{min} = c_{swap}$ **then**
      Schedule and perform swaps, then update qubit positions Q
    else if $c_{min} = c_{switch}$ **then**
      Perform quantum switch interaction directly without moving qubits
    else
      Perform appropriate teleportation operation
    end if
  end for
  Update gate list Ω with executed operations
  return Ω
end procedure



(a) (b)

Fig. 4. Five-qubit QAOA-Maxcut circuit example. (a) Input problem graph. (b) Valid circuit.

experiments, we compare our linear compiler [11] with the SABRE compiler, focusing on gate count and circuit depth. For the quantum-switch integration experiments, we use a simulation framework that models quantum circuit compilation within a DQC setting with realistic performance metrics derived from IBM's latest quantum processors and theoretical quantum-switch implementations.

The architecture consists of superconducting qubits arranged in a heavy-hex lattice, split evenly across multiple QPUs. Communication between these QPUs is facilitated through dangling qubits serving as interconnects. We use QAOA and QFT as our benchmark circuits, with total qubit counts varying from 10 to 200 for the baseline experiments and fixed at 400 for the extended experiments focused on quantum-switch configurations and performance metrics.

We use single-layer QAOA and standard QFT circuits as benchmarks. We generate random regular graphs for QAOA with degree = 0.4× qubits number. This ensures each qubit connects to approximately 40% of other qubits, creating moderate connectivity that avoids both sparse and overly dense

TABLE II
COMPARISON OF GATE COUNT AND CIRCUIT DEPTH BETWEEN SABRE AND LINEAR COMPILER FOR QAOA CIRCUITS

Qubit Count	Gate Count		Circuit Depth	
	SABRE	Linear	SABRE	Linear
10	408	175	200	83
20	1631	730	625	193
50	9895	4675	2136	523
80	26149	12040	3819	853
100	40837	18850	5031	1073
120	59923	27180	6097	1293

TABLE III
COMPARISON OF GATE COUNT AND CIRCUIT DEPTH BETWEEN SABRE AND LINEAR COMPILER FOR QFT CIRCUITS

Qubit Count	Gate Count		Circuit Depth	
	SABRE	Linear	SABRE	Linear
10	109	107	81	59
20	523	487	407	126
50	4176	3076	2577	376
80	11676	8164	6472	617
100	20664	12854	10849	784
120	30479	18671	15959	936

special. A sample QAOA circuit and problem graph are shown in Fig 4. QFT circuits follow a standard recursive structure with controlled rotations and final Hadamard operations.

We compare against SABRE compiler to demonstrate the fundamental advantages of LNN topology transformation from heavy-hex structures, establishing the foundation upon which our distributed optimizations build. SABRE represents the state-of-the-art for heavy-hex compilation and provides a realistic baseline for measuring the benefits of topology transformation. For DQC scenarios, direct comparison with single-QPU LNN compilers [11], [38] is not applicable because these methods cannot address inter-QPU communication requirements or quantum-switch utilization decisions that are central to our EDDQC approach.

B. Part 1: LNN-DQC Feasibility in Ideal Environment

In this first part of our evaluation, we demonstrate the feasibility and advantages of LNN-DQC in an ideal environment, focusing on the fundamental improvements provided by our linear compilation approach and dangling-qubit utilization.

1) Experiment 1: Comparison Between SABRE and Linear Compiler: This experiment compares the performance of the traditional SABRE compiler with our linear compiler. Key performance metrics, including gate count and circuit depth, are evaluated across various QAOA and QFT circuits, as the run-time advantage of our linear compiler has already been demonstrated in previous work [11], [38].

The results, as summarized in Tables II and III, show clear advantages for the linear compiler. For instance, in a 100-qubit QFT circuit, the linear compiler reduces the gate count by 48%, from 148398 (SABRE) to 76795 gates, and decreases the circuit depth by 25%.

These improvements stem from LNN's efficient qubit placement, minimizing SWAP gates for nonlocal interactions.

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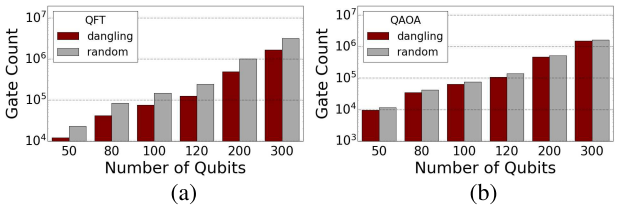


Fig. 5. Gate count comparison between random and dangling-qubit placement for QFT and QAOA circuits. (a) Gate count of QFT. (b) Gate count of QAOA.

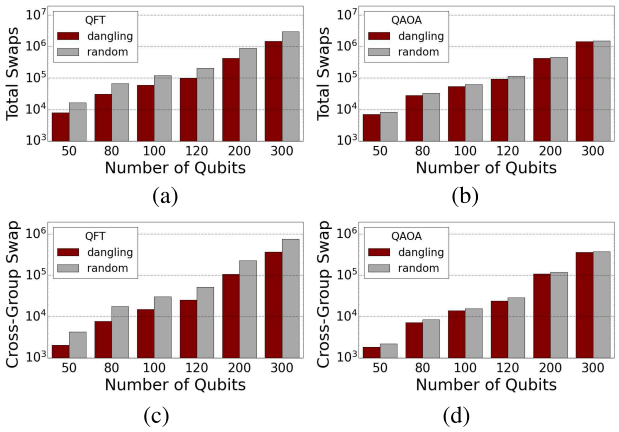


Fig. 6. Total SWAP gates and cross-group SWAP gates comparison between random and dangling-qubit placement. (a) QFT SWAP. (b) QAOA SWAP. (c) QFT cross-group SWAP. (d) QAOA cross-group SWAP.

SABRE excels in smaller circuits but faces challenges with larger ones due to hardware topology constraints, leading to higher gate counts and deeper circuits.

2) Experiment 2: Nonlocal Gate Placement With Dangling Qubits: In this experiment, we introduce a new metric: cross-group SWAP, measuring qubits that traverse nonlocal gates via SWAP operations during compilation. This metric is crucial in DQC, affecting both circuit complexity and fidelity, with further discussion in Section VI.

For each QPU, we randomly select 20% of qubits to match the proportion of dangling qubits and serve as communication interfaces. These randomly selected qubits are distributed throughout the LNN chain rather than positioned at optimal endpoints. The random selection uses a uniform probability distribution across all qubit positions and is repeated across multiple trials to ensure statistical validity.

As shown in Fig. 5, the dangling-qubit method consistently reduces gate count across all qubit sizes, nearly halving the gate count in a 100-qubit circuit compared with the random method. Similarly, Fig. 6 shows a significant reduction in both total and cross-group SWAP gates, with the 100-qubit circuit seeing a drop from 15117 to 7216 cross-group SWAPs.

The considerable reduction in both gate count and SWAP gates highlights the superiority of the dangling-qubit method, especially for larger circuits. This reduction is critical for minimizing overhead in distributed quantum systems, improving both scalability and circuit fidelity.

C. Part 2: Enhanced Performance With Realistic Metrics

In this second part of our evaluation, we extend our analysis to include realistic performance metrics and examine the benefits of quantum-switch integration in practical scenarios.

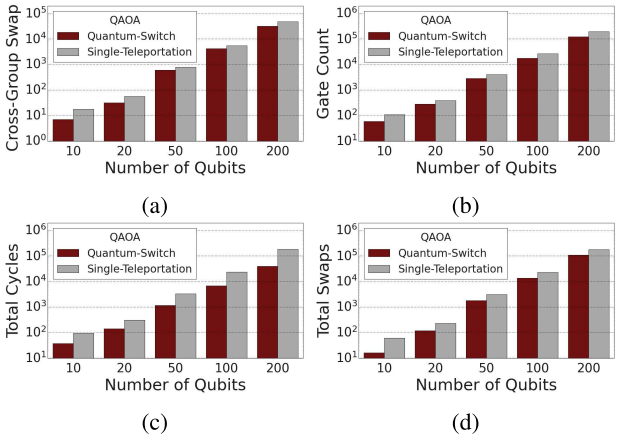


Fig. 7. Detailed performancecomparison of QAOA circuits between quantum-switch DQC and single-teleportation DQC, ranging from 10 to 200 qubits. (a) QAOA cross-group SWAP. (b) QAOA gate count. (c) QAOA cycle count. (d) QAOA SWAP.

These experiments provide guidance for experimental research by identifying optimal configurations and tradeoffs in DQC implementations. We do not use default values for "max parallel count" and "quantum-switch count" as these are the key optimization variables in our study. Max parallel count varies from 2 to 20, and quantum-switch count varies from 1 to 20 across different experiments. When one parameter is swept, others are held at optimal values determined for that specific analysis.

The computational tradeoff of 2-D parameter sweeps becomes significant when considering simulation complexity. Such sweeps would increase evaluation requirements from $O(M + S) = 39$ evaluations to $O(M \times S) = 380$ evaluations, representing a 9.7× increase in computational cost while providing minimal additional optimization benefit due to the demonstrated parameter independence.

Our experimental results confirm this separable optimization behavior, with optimal parameters for QAOA: $(m^*, s^*) = (10, 8)$ and QFT: $(m^*, s^*) = (8, 5)$ achievable through independent single-parameter optimization. This validation supports our methodological choice to prioritize computational efficiency without sacrificing optimization quality.

While comprehensive 2-D parameter exploration remains a valuable direction for future research, particularly for larger system scales, such studies would require dedicated computational resources. Our current approach provides practical optimization guidance while maintaining computational feasibility for comprehensive algorithm evaluation.

1) Experiment 3: Performance Comparison Between Quantum-Switch DQC and LNN-DQC: The goal of this experiment was to compare the effectiveness of quantum-switch-enabled DQC against a traditional LNN-DQC under performance constraints. Both configurations were subjected to identical interaction graphs to ensure a fair comparison, with performance metrics derived from IBM's latest quantum processors and quantum-switch implementations.

The results, presented in Figs. 7 and 8, indicate that the quantum-switch DQC consistently outperforms the LNN-DQC. Specifically, in QAOA circuits, quantum-switch DQC demonstrates significantly fewer cross-group swaps (35%

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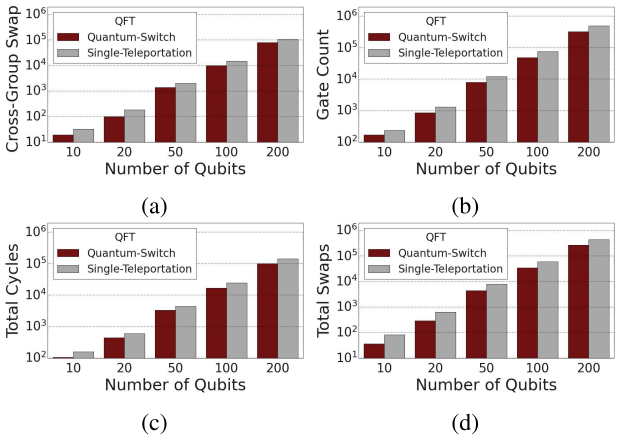


Fig. 8. Detailed performance comparison of QFT circuits between quantum-switch DQC and single-teleportation DQC, ranging from 10 to 200 qubits. (a) QFT cross-group SWAP. (b) QFT gate count. (c) QFT cycle count. (d) QFT SWAP.

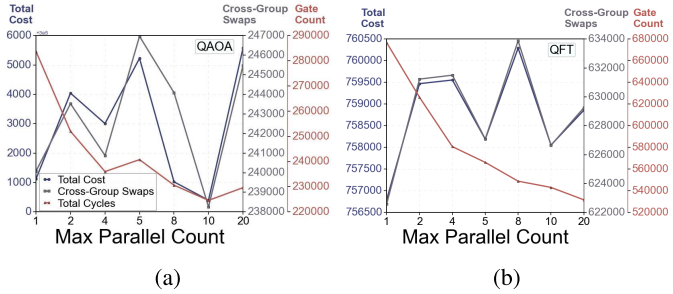


Fig. 9. Evaluation of max parallel metrics. The graphs illustrate how varying the number of maximum parallel entangled qubit pairs impacts performance for QAOA and QFT circuits. (a) QAOA max parallel count. (b) QFT max parallel count.

reduction on average), a lower total gate count (20% reduction), and reduced total cycles (up to 40% fewer), which translates to minimized qubit movement overhead. The QFT circuits exhibit a similar trend, with up to a 45% reduction in cross-group swaps and a 25% decrease in gate count compared with the single-teleportation approach.

These results substantiate the hypothesis that the integration of quantum switches markedly enhances computational efficiency in DQC systems. By reducing the need for physical qubit movement through strategic entanglement, we achieve more efficient circuit execution with lower error rates.

2) Experiment 4: Impact of Maximum Entangled Qubits per Quantum Switch: To provide practical guidance for experimental research, we investigated how the maximum number of qubits a quantum switch can entangle simultaneously affects system performance. We varied the entanglement capacity (max parallel) from 2 to 20 qubits while fixing the total qubit count at 400, focusing on realistic metrics including time, error rates, and total execution cost.

Results indicate that for QAOA circuits, the total cost experiences significant increases at max parallel values of 5 and 20, while QFT circuits show inefficiencies at 8 and 20, as illustrated in Fig. 9. Cross-group swaps and total cycles tend to decrease as max parallel increases, hitting a near-optimal point around 10 but rising again at extreme values.

A deeper analysis reveals that a max parallel value of 10 strikes an efficient balance for QAOA, reducing cycles

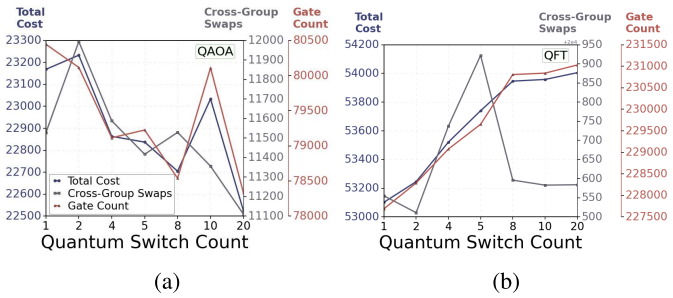


Fig. 10. Evaluation of quantum-switch metrics. These graphs compare the performance impact of varying the number of quantum switches for QAOA and QFT circuits. (a) QAOA quantum switch count. (b) QFT quantum switch count.

while keeping total costs manageable. In contrast, QFT circuits perform more efficiently with max parallel values of 5 or 8, minimizing total cycles and keeping gate counts under control. These findings provide valuable guidance for experimental research by identifying the optimal entanglement capacity for different quantum algorithms, helping researchers focus their efforts on the most promising configurations.

3) Experiment 5: Effect of the Number of Quantum Switches on Performance: This experiment explored the impact of varying the number of quantum switches, with counts ranging from 1 to 20, using realistic performance metrics. The results for QAOA circuits show a consistent decrease in total cost as the quantum-switch count increases, with the lowest costs achieved at 20 switches. However, QFT circuits present a different behavior: lower switch counts (1 or 2) tend to minimize costs, while higher counts inflate the gate count and overall complexity. These results are depicted in Fig. 10.

Cross-group swaps in QAOA circuits stabilize as switch counts increase, with early inefficiencies peaking around a count of 2. For QFT, swaps spike at five switches before leveling off. Gate counts in QAOA are optimized with higher switch numbers, while QFT circuits benefit from moderate values, indicating a nuanced dependency on the circuit structure.

These results suggest that QAOA circuits are best served by a switch count of 8–10, while QFT circuits achieve optimal performance with 5–8 switches. This insight is valuable for experimental researchers designing DQC platforms, as it allows for algorithm-specific optimization of the network topology and helps focus research efforts on the most promising architectural configurations.

D. Summary of Evaluation Results

The linear compiler itself provides substantial improvements over SABRE, while the addition of quantum switches further enhances performance by reducing cross-group operations and minimizing qubit movement. Our HLC-SFC algorithm family definition provides the theoretical foundation for understanding which quantum algorithms benefit most from LNN-based distributed compilation. Our analysis identifies critical configuration thresholds that optimize performance. Specifically, a maximum entanglement capacity of M = 5–10 and a quantum-switch count of N switch =5–8 yield the best performance balance for most applications. These parameters minimize the total execution cost and optimize resource utilization without introducing unnecessary complexity or overhead.

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The significant reduction in cross-group swaps (up to 67.2%), gate count (up to 43.8%), and total cycles (up to 40%) demonstrates the substantial efficiency gains possible through our combined approach. These improvements validate that HLC-SFC algorithms naturally align with LNN topology benefits because their predominantly local interactions allow efficient decomposition of long-range connectivity requirements. These improvements translate directly to lower error rates and better scalability for large-scale quantum computations, addressing two of the primary challenges in DQC systems.

Our evaluation focuses on single-layer QAOA and QFT as representative benchmarks spanning the quantum computing spectrum. QAOA represents NISQ optimization algorithms, while QFT serves as the foundational building block for fault-tolerant algorithms, including Shor's factoring and quantum arithmetic. This selection covers the most significant quantum algorithms from near-term to fault-tolerant systems, with future work expanding to additional HLC-SFC algorithms.

Previous research demonstrates LNN compilation benefits for various quantum algorithms [11], [38], [39], and our distributed extensions inherit these advantages while addressing scalability challenges. Algorithms with local connectivity pattern, like quantum simulation and variational eigensolvers, are particularly well-suited. Performance improvement depends on circuit structure, algorithms requiring extensive long-range connectivity see diminished benefits, while HLC-SFC pattern algorithms achieve optimal performance.

Large-scale DQC faces potential error rate increases with physical distance between dangling qubits. Contemporary quantum networking research addresses distance-related fidelity degradation through improved cavity coupling and enhanced photonic interfaces [33], [36]. Our EDDQC approach mitigates these concerns by minimizing inter-QPU communication through strategic LNN topology, keeping most operations local within individual QPUs.

Furthermore, our results provide valuable guidance for experimental research by identifying the optimal configurations for different quantum algorithms and highlighting the tradeoffs involved in DQC implementations. While LNN topology represents a theoretical optimization that faces current hardware fabrication challenges, our heavy-hex transformation approach demonstrates how existing quantum infrastructure can achieve near-optimal distributed computing performance. These insights will help focus research efforts on the most promising architectural designs and accelerate the development of practical, large-scale quantum computing systems.

VI. Conclusion and Discussion

Our EDDQC approach represents a significant advancement in DQC, combining the computational efficiency of LNN chains with the flexibility of reconfigurable quantum networks. By transforming the conventional heavy-hex topology into an LNN structure with dangling qubits at chain endpoints, we create ideal communication links between QPUs that leverage quantum switches to establish a dynamically reconfigurable interconnect topology. This strategic utilization of dangling qubits aligns with fundamental theoretical principles, particularly the area law of entanglement entropy in 1-D systems, ensuring efficient scaling of quantum resources as system size increases. While our current experiments use simulators due to ongoing implementation of nonlocal gates in real-quantum computers, our method's ability to minimize cross-group operations will be particularly valuable in hardware environments where nonlocal gates typically have lower fidelity. Our design primarily uses a single pair of dangling qubits per quantum switch, but the LNN architecture can support more complex configurations, opening opportunities for further optimization by aligning dangling-qubit connections with specific algorithm requirements.

Acknowledgment

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